KAIZEN MATHS

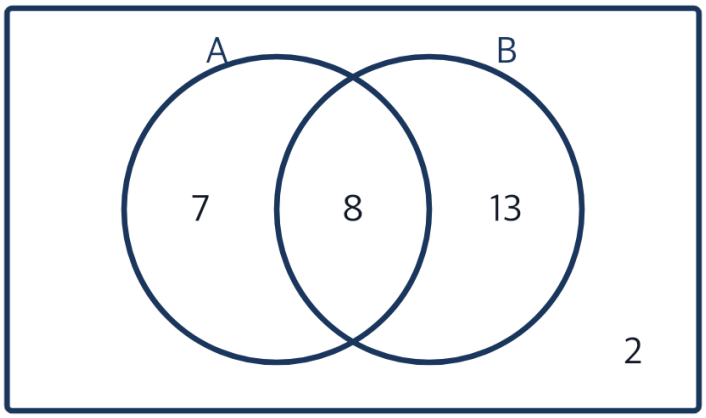
Lesson starter

Venn Diagrams and Set Probability

Two-Set Diagrams

Name _____

SA1



Use the Venn diagram to find **n(B)**.

STA-B8CE-L1 · SA

Show your working

KAIZEN MATHS

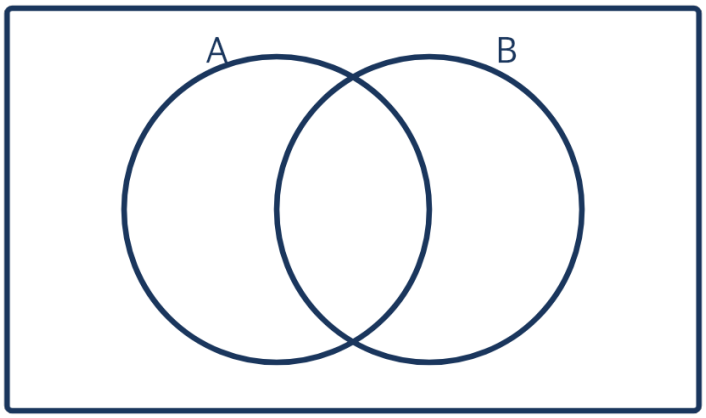
Lesson starter

Venn Diagrams and Set Probability

Two-Set Diagrams

Name _____

SB1



In a class of 61, n(A)=30, n(B)=18, and n(A ∩ B)=5. Complete the Venn diagram and find how many are in neither set.

STA-B8CE-L1 · SB

Show your working

KAIZEN MATHS

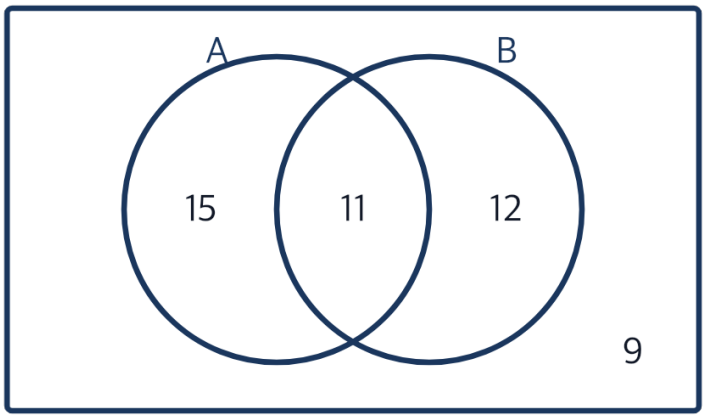
Lesson starter

Venn Diagrams and Set Probability

Two-Set Diagrams

Name _____

SC1



Use the Venn diagram to find **n(A ∪ B)**.

Show your working

KAIZEN MATHS

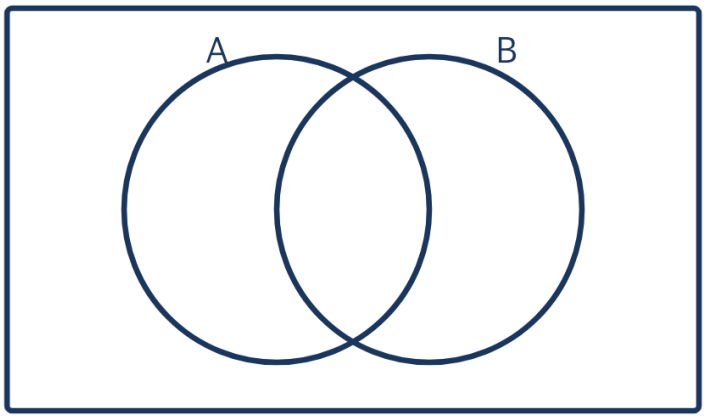
Lesson starter

Venn Diagrams and Set Probability

Two-Set Diagrams

Name _____

SD1



In a class of 53, n(A)=29, n(B)=25, and n(A ∩ B)=9. Complete the Venn diagram and find how many are in neither set.

Show your working

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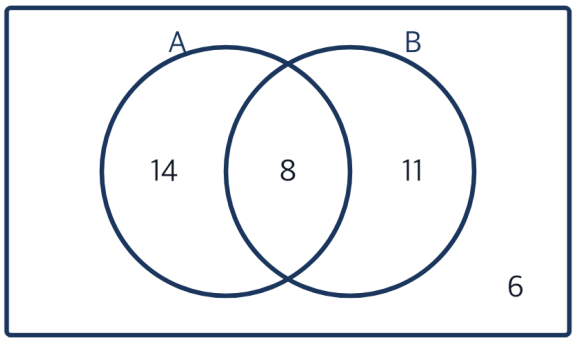
Exit ticket

Venn Diagrams and Set Probability

Two-Set Diagrams

Name _____

EA1



Use the Venn diagram to find **n(A)**.

Show your working

KAIZEN MATHS

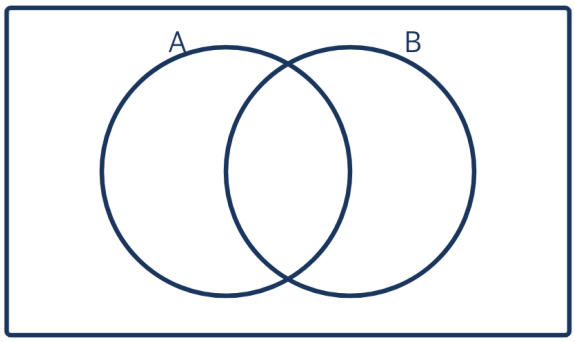
Exit ticket

Venn Diagrams and Set Probability

Two-Set Diagrams

Name _____

EB1



In a class of 71, n(A)=30, n(B)=35, and n(A ∩ B)=6. Complete the Venn diagram and find how many are in neither set.

Show your working

KAIZEN MATHS

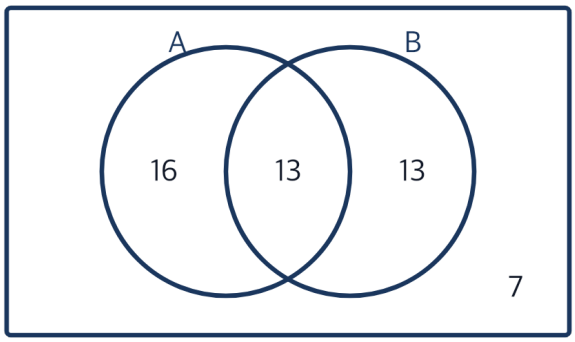
Exit ticket

Venn Diagrams and Set Probability

Two-Set Diagrams

Name _____

EC1



Use the Venn diagram to find **n(A ∩ B)**.

Show your working

KAIZEN MATHS

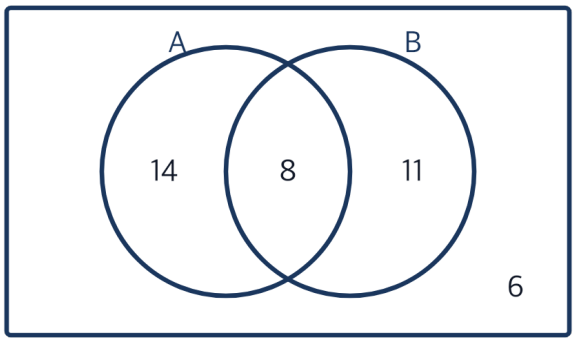
Exit ticket

Venn Diagrams and Set Probability

Two-Set Diagrams

Name _____

EA1



Use the Venn diagram to find **n(A)**.

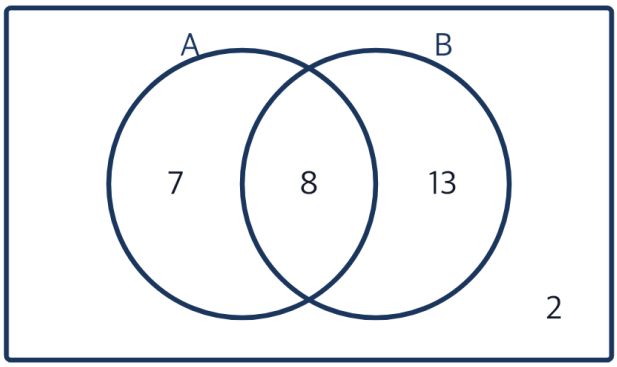
Show your working

Teacher answers & working

Venn Diagrams and Set Probability · Two-Set Diagrams

STA-B8CE-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Use the Venn diagram to find **n(B)**.

Answer:

21

Working:

n(B) means everything inside circle B.

$n(B) = n(A' \cap B) + n(A \cap B)$

This means add B only and the overlap.

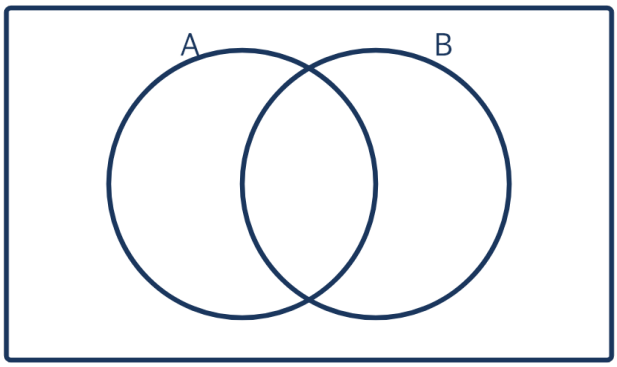
$n(B) = 13 + 8 = 21$

Teacher answers & working

Venn Diagrams and Set Probability · Two-Set Diagrams

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SB1



In a class of 61, n(A)=30, n(B)=18, and n(A ∩ B)=5. Complete the Venn diagram and find how many are in neither set.

Answer:

Neither = 18

Working:

Start with the overlap: n(A ∩ B) = 5.

$n(A \cap B') = n(A) - n(A \cap B) = 30 - 5 = 25.$

This is A only: in A but not in B.

$n(A' \cap B) = n(B) - n(A \cap B) = 18 - 5 = 13.$

This is B only: in B but not in A.

n((A ∪ B)') = total - n(A ∪ B).

This means neither A nor B.

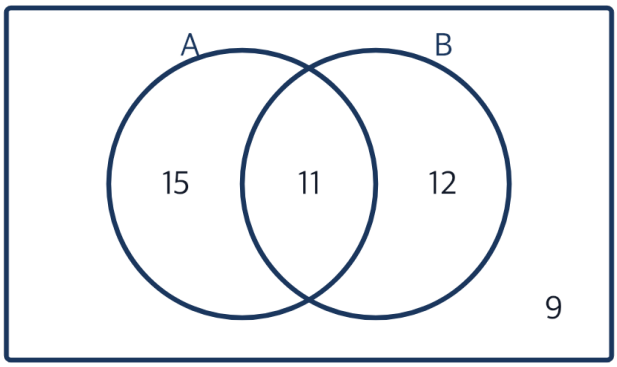
$n((A \cup B)') = 61 - (25 + 5 + 13) = 18.$

Teacher answers & working

Venn Diagrams and Set Probability · Two-Set Diagrams

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SC1



Use the Venn diagram to find **n(A ∪ B)**.

Answer:

38

Working:

A ∪ B means inside A or inside B or both.

$n(A \cup B) = n(A \cap B') + n(A \cap B) + n(A' \cap B)$

This means A only, plus the overlap, plus B only.

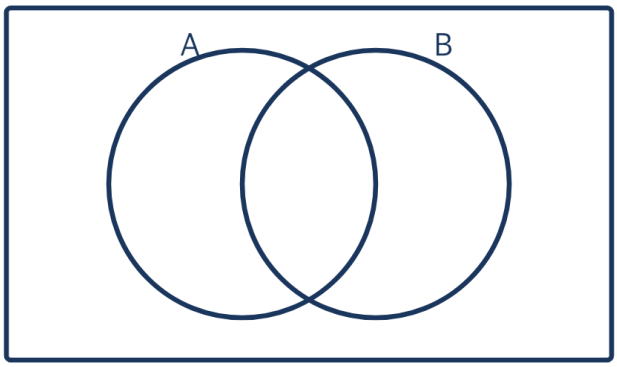
$n(A \cup B) = 15 + 11 + 12 = 38$

Teacher answers & working

Venn Diagrams and Set Probability · Two-Set Diagrams

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SD1



In a class of 53, n(A)=29, n(B)=25, and n(A ∩ B)=9. Complete the Venn diagram and find how many are in neither set.

Answer:

Neither = 8

Working:

Start with the overlap: n(A ∩ B) = 9.

$n(A \cap B') = n(A) - n(A \cap B) = 29 - 9 = 20.$

This is A only: in A but not in B.

$n(A' \cap B) = n(B) - n(A \cap B) = 25 - 9 = 16.$

This is B only: in B but not in A.

$n((A \cup B)')$ = total - n(A ∪ B).

This means neither A nor B.

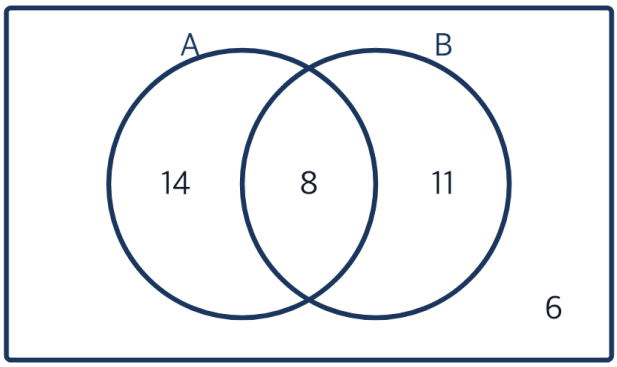
$n((A \cup B)') = 53 - (20 + 9 + 16) = 8.$

Teacher answers & working

Venn Diagrams and Set Probability · Two-Set Diagrams

STA-B8CE-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



Use the Venn diagram to find **n(A)**.

Answer:

22

Working:

n(A) means everything inside circle A.

$n(A) = n(A \cap B') + n(A \cap B)$

This means add A only and the overlap.

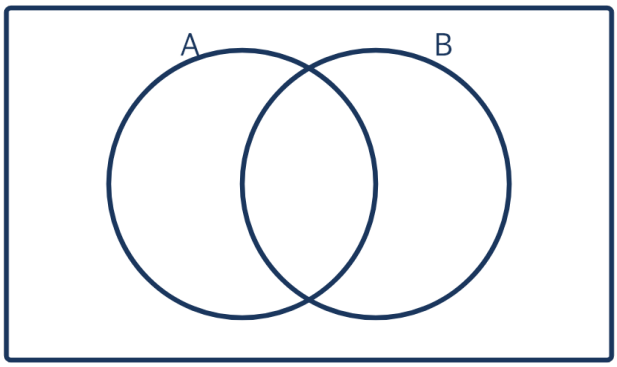
$n(A) = 14 + 8 = 22$

Teacher answers & working

Venn Diagrams and Set Probability · Two-Set Diagrams

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EB1



In a class of 71, n(A)=30, n(B)=35, and n(A ∩ B)=6. Complete the Venn diagram and find how many are in neither set.

Answer:

Neither = 12

Working:

Start with the overlap: n(A ∩ B) = 6.

$n(A \cap B') = n(A) - n(A \cap B) = 30 - 6 = 24.$

This is A only: in A but not in B.

$n(A' \cap B) = n(B) - n(A \cap B) = 35 - 6 = 29.$

This is B only: in B but not in A.

n((A ∪ B)') = total - n(A ∪ B).

This means neither A nor B.

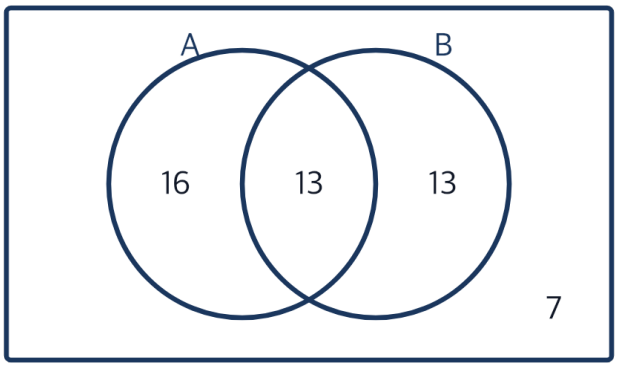
$n((A \cup B)') = 71 - (24 + 6 + 29) = 12.$

Teacher answers & working

Venn Diagrams and Set Probability · Two-Set Diagrams

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EC1



Use the Venn diagram to find **n(A ∩ B)**.

Answer:

13

Working:

A ∩ B means the overlap of A and B.

$n(A \cap B) = 13$.